

(This command will create a new directory with the command `mkdir` as the user Mary with Mary's current group set to David. Mary, however, must be a member of the David group.)

- `Sudo -v`

(It resets or extends the timeout for automated authentication, allowing you to continue by using the `sudo` commands, without submitting the password.)

- `Sudo -k`

(It will terminate the `sudo` authentication of the existing user. Therefore, the next `sudo` command will need password verification.)

Dealing with file permissions through the command line

There are two simple commands to manipulate the file ownership and permissions in the Linux file system as depicted below:

For changing ownerships, you can use the command: `chown`

Likewise, to change the permission of a file, you can use the command: `chmod`

It is not difficult to use either of these two commands. However, note that only the root user or the current owner can change permissions or file ownership.

For example, a new folder has been created with the name `/DAT/SHA` in the data partition. Users Mary and David need to read and write access to this folder. There are several ways to do this. If Bethany and Jacob are the only users of the system (and the network is known to be very safe), you can change the folder permissions to allow access. One way to do this is to issue the command:

- `sudo chmod -R ugo+rw /DAT/SHA`

This command is explained in detail as follows-

`sudo` - It renders admin privileges to the executed command (otherwise the command given above must be rooted and run without "sudo")

`Chmod`-command to change permissions

`-R`- It changes the permissions of the parent folder and its dependent objects

`ugo + rw`- Here users, group, and others are granted read-write permissions

`/DAT/SHA`- indicates the modified directory

As you can see, this command opens the `SHARE` folder and makes it accessible to anyone in the system. It must be remembered that this is purely for demonstration. As far as, the permission part is concerned, it is explained below:

- `u` for user
- `g` for group
- `o` for others